Switching Circuits and Switching Expressions

Two Types of Switches

Normally Open Switch

Normally Closed Switch

Two Types of Connections

Series Connection

Parallel Connection

EX-OR

Majority of 3

F(x,y,z) = 1 when two or more inputs are 1's

0 otherwise

Series-Parallel Switching Circuits

- Building Blocks
  - Normally open switch
  - Normally closed switch

- Composition Rules
  - Series
  - Parallel

Are these primitives sufficient to implement any Boolean function?

Synthesis of Series-Parallel Switching Circuits

- From Truth Tables

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Synthesis of Series-Parallel Switching Circuits

- From Expressions

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

- Canonical Sum-of-Products (SOP) Expressions

F = A’BC’ + A’BC + AB’C’ + AB’C + ABC

- Minimal SOP Expressions, usually obtained by the K-map Method

A \ B,C	00	01	11	10
0	0	0	1	1
1	1	1	1	0

F = AB’ + BC + A’B

- Nested Expressions

F = AB’ + B(C + A’)

Nested AND-OR Expressions

Any nested AND-OR expression of positive and negative variables can be implemented as a series-parallel switching network.

Given variables x_1, x_2, …

- x_i is an expression
- x_i' is an expression
- If F and G are expressions, then
- (F.G) is an expression
- (F+G) is an expression

Notes:
Parentheses can be omitted when there is no confusion.

. has presedence over +

. may be omitted when there is no confusion

Examples

F = ((a’.b)+(c.(a’+d’)))
F = a’b + c(a’+d’)

G = (xyz + y’z + y)u

H = (a.b).(c.d)+c’d

General Boolean Expressions

General nested Boolean AND-OR-NOT expressions can be implemented after using the De Morgan's Laws to move all negations from subexpressions to variables.

De Morgan's Laws

$(F+G+\ldots)' = F'.G'\ldots.$

$(F.G\ldots)' = F' + G' + \ldots$

Example

Consider $F = AB' + B(C + A')$

Suppose we wish to implement F' as a series-parallel switching circuit.

$$F' = (AB' + B(C + A'))'$$

$$= (A'+B).(B' + C'.A)$$

$$= A'B' + ABC'$$

Subcircuit Connection

Switch as a 3-terminal Device

Primitives

- Building Blocks
  - Normally open switch
  - Normally closed switch

- Composition Rules
  - Series
  - Parallel
  - Subcircuit

Sequential Circuits – Need Memory

Example: Dating Machine

On any given day, Jack and Jill can go on a date provided

1) they both agree to go on a date and
2) they did not go on a date on the day before

Primitives

- Building Blocks
  - Normally open switch
  - Normally closed switch

- Composition Rules
  - Series
  - Parallel
  - Subcircuit

- Memory

Implementation of switching circuits in a physical medium

Any medium which supports the primitives:

- Electromagnetic
- Hydraulic
- Pneumatic
- Solid-state electronic
  - Bipolar
  - Field-effect
- …..

Physical Characteristics

- Space
- Speed
- Power
- Reliability
- Cost
- ….

CMOS:
Complementary Metal-Oxide Silicon